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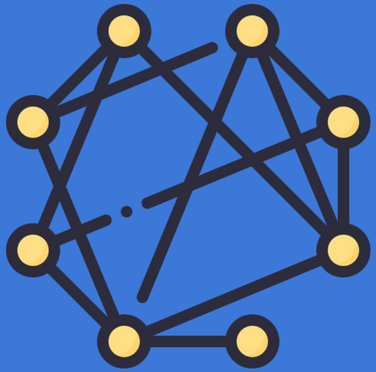

Antonym and Synonym Distinction using Graph based Neural Network

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Overview

"He is not ____. Indeed it's better to be ____ than ____"

Antonymy - Synonymy

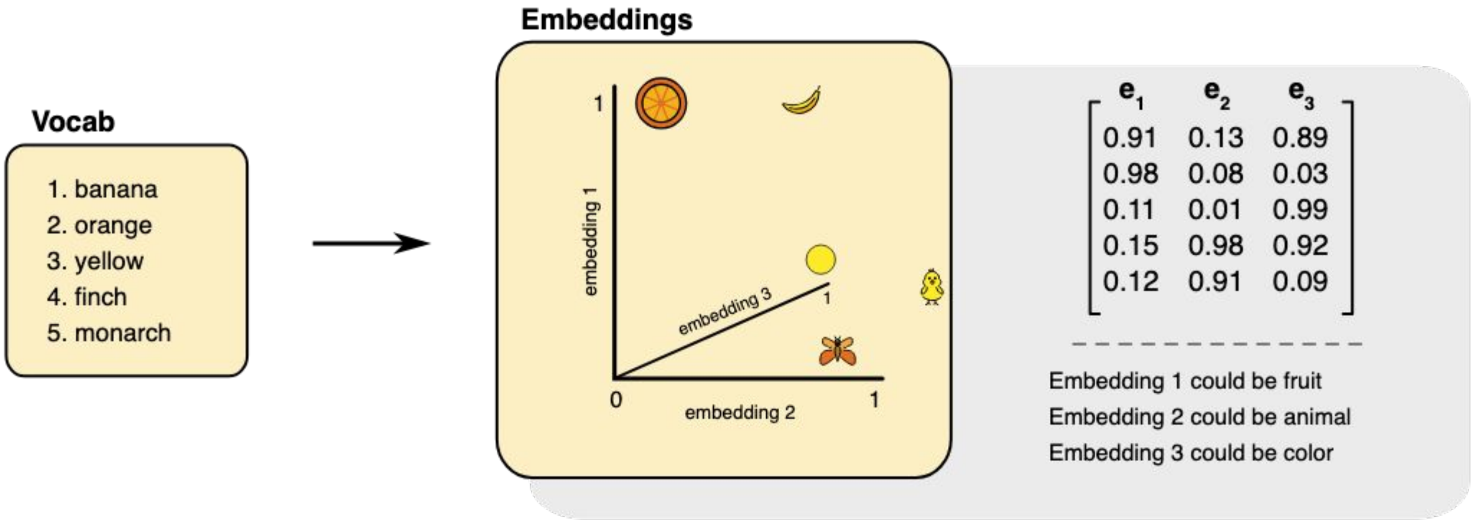
- Synonyms share similar meanings, allowing for substitution in many contexts.
- Antonyms express direct opposition, serving to contrast or negate meanings.
- Understanding and distinguishing these relationships enhances an NLP system's ability to interpret and manipulate language effectively.
- Some applications include sentiment analysis, text summarization & paraphrasing, and language generation.
- The distributional hypothesis states that words with similar meanings will have similar distributional properties.
- NLP models naturally group synonyms together based on their usage patterns in large corpora.

Handling antonymy is a challenging task in NLP

- Antonyms often appear in similar or even identical contexts as synonyms.
- This contextual overlap means that simple co-occurrence based embeddings may not effectively distinguish them, as both can have high vector similarity.
- Antonymy can be implicit, requiring a deeper understanding of negation, contrast, and world knowledge.
- It is difficult to discern the correct sense without context (*dull* or *dim* for *bright*).
- Good quality datasets are scarce, especially for low-resource languages.

Background Literature

- Static word embeddings (e.g., Word2Vec, FastText, GloVe) generate fixed vector representations, ignoring context.



Background Literature

- Pattern-based neural models leverage lexico-syntactic structures and morphological cues.
  - AntSynNET hypothesizes that antonym pairs co-occur in specific syntactic structures (e.g., "either X or Y") more frequently than synonyms.
  - AntNet extracts dependency paths between word pairs, focusing on both the structure and the morphological features of the words.
  - Challenges include the requirement of extensive labeled corpora, noise sensitivity, and domain specificity.

Background Literature

- Subspace distillation enhances static embeddings by isolating synonymy and antonymy into distinct subspaces to model their unique algebraic properties.
- Thesauri-Augmented Embeddings combines distributional signals (from corpora) with explicit relational knowledge.
- They struggle with words that have multiple senses and do not work for rare or unseen words.

Background Literature

- Contextual embeddings generate unique representations for the same word depending on its context.
  - BERT uses bidirectional attention to analyze both left and right context of a word.
  - ELMo combines forward and backward LSTM layers to encode sentence context.
  - They handle polysemy via dynamic representations.
  - Can be fine-tuned for downstream tasks.

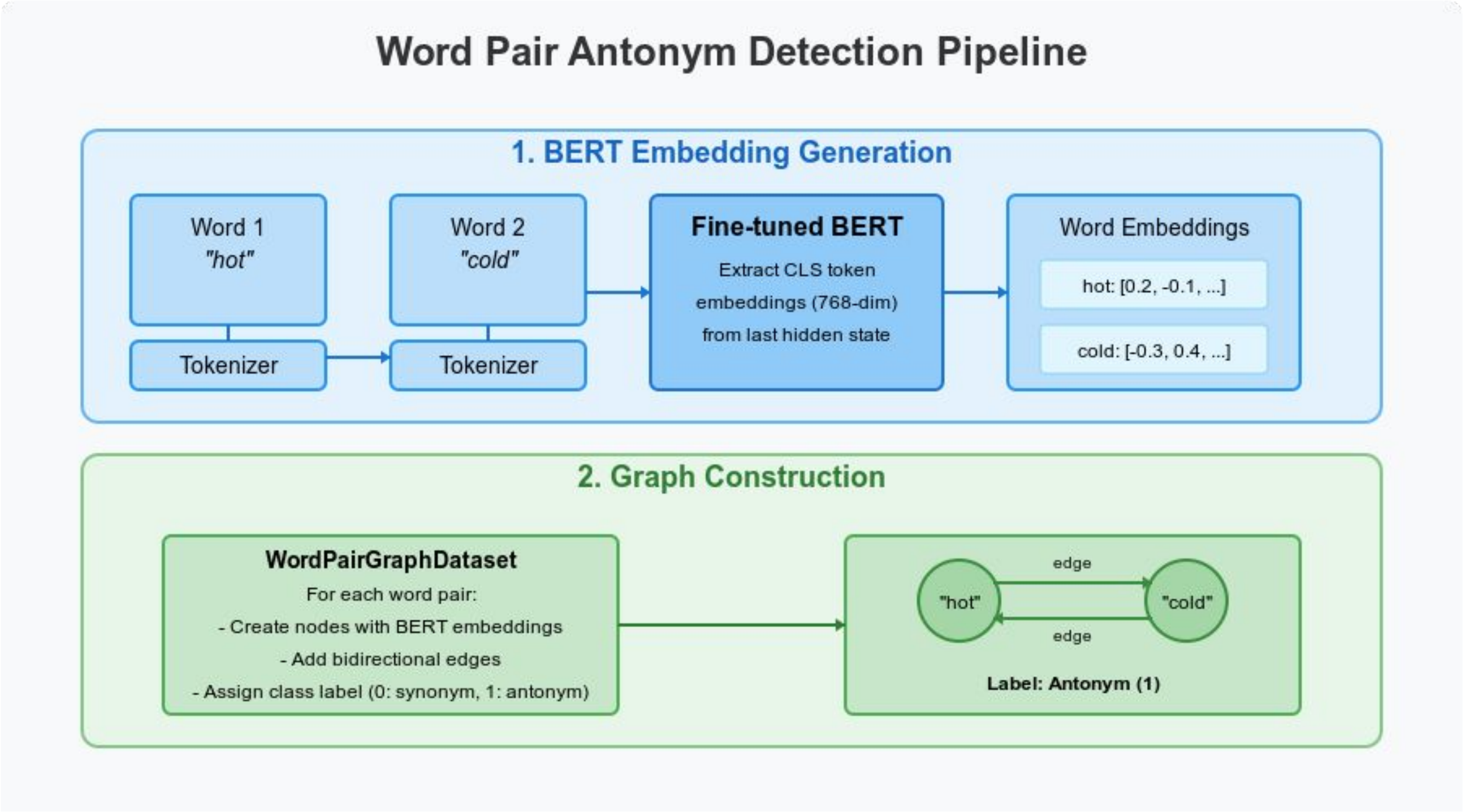
ICE-NET (InterlaCed Encoder NETworks), 2024

- ICE-NET distinguish antonyms from synonyms by explicitly modeling their unique relational properties : symmetry, transitivity, and trans-transitivity.
- It takes pre-trained word embeddings as input that capture semantic information about words from large text corpora.
- Use multiple encoders to model r
- elation-specific properties.
- Employ attentive graph convolutions to share information across related word pairs.
- Determine edge weights by attention mechanisms to prioritize influential neighbors during information propagation.

Methodology

- The aim is to model the understanding of antonyms and synonyms over the vector embeddings space.
- The transformer architecture is large and has good syntactic knowledge that we can leverage.
- We analyzed the activations of neurons of BERT and found that most of the attentive weights with maximum activation delta and usage were generally (about 75%) transitively aligned, implying some underlying graph-like propagation.
- We fine-tuned a BERT uncased model, leveraging its embeddings to create initial graph nodes.
- These nodes are then used as the base for a GCN/GAT to train and test on the benchmark dataset by Nguyen et al. (2017).

Methodology



Dataset Used

- Used the dataset by Nguyen et al. (2017) that was manually curated from WordNet (lexical database) and Wordnik (online dictionary).
- It comprises randomly split synonym and antonym pairs corresponding to three word classes (adjective, noun, and verb).
- 1:1 ratio of antonyms to synonyms within each word class.
- Pairs are annotated by experts for correctness and relevance.

word class	train	dev	test
Adjective	5562	398	1986
Noun	2836	206	1020
Verb	2534	182	908

```
thick	stocky	0
unstressed	stressed	1
miraculous	natural	1
honest	incorruptible	0
fourth	4th	0
immaterial	material	1
conventional	unconventional	1
confusing	clear	1
```

Architectures and results

We experimented with 5 different architectures involving graph convolution and attention and multi-headed attention networks involving k-fold validations, contrastive learning parameters, and other neat tricks.

Model	Overall Test Accuracy
Multihead Attention	0.86
Single Conv Graph	0.87
Contrastive GAT	0.89
Multiscale GAT	0.90
Dual Encoder	0.91

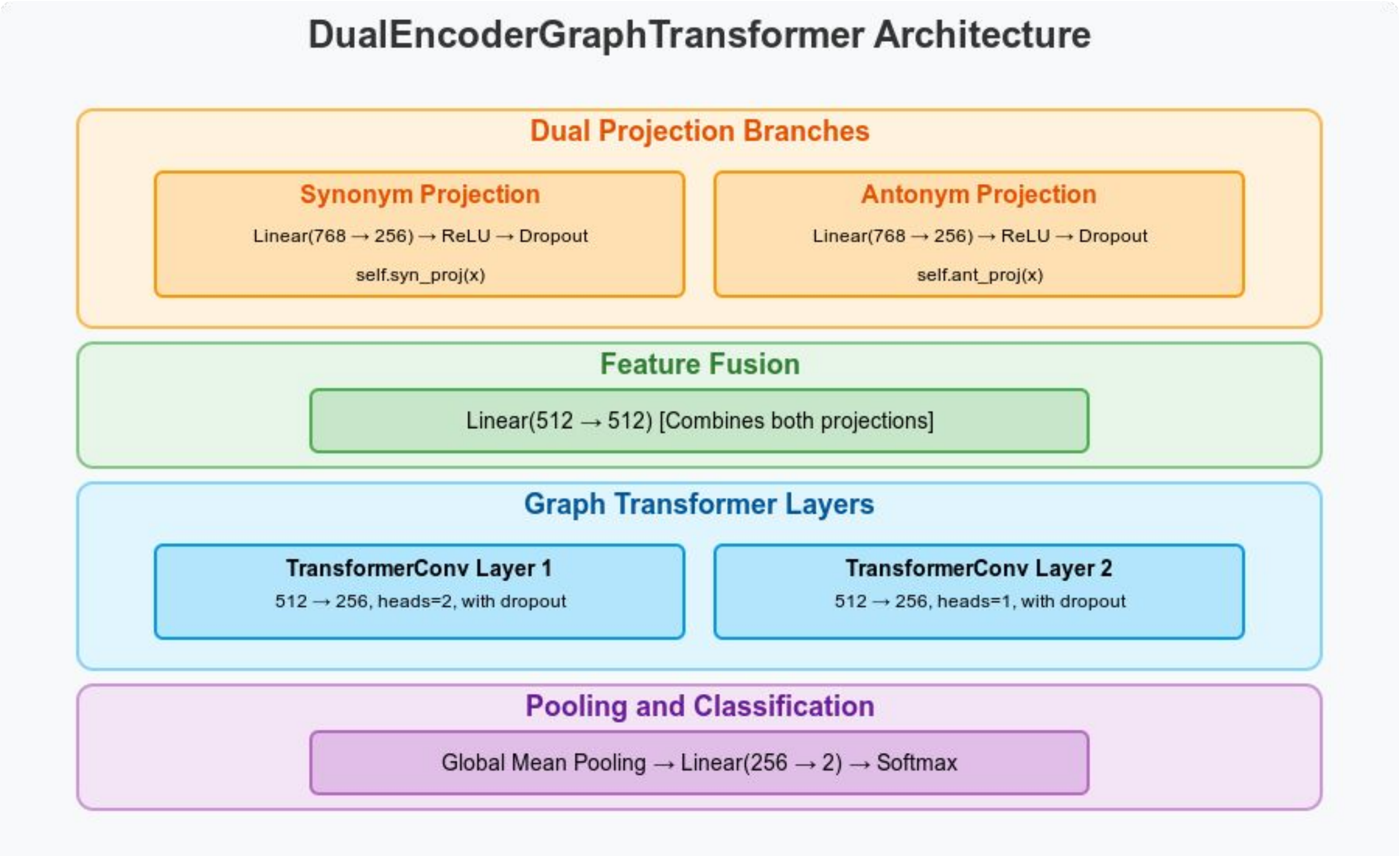
Best Architecture

The dual encoder convolution method yielded the best results.

Dual Encoder Projections: Inputs are passed through two separate branches — one for synonyms and one for antonyms—each projecting into its own semantic space.

- Fusion Layer: Outputs from both branches are concatenated and passed through a linear layer to fuse semantic signals before graph processing.
- Graph Transformer Layers: A stack of TransformerConv layers captures interactions between word pairs as small graphs, leveraging attention for relation modeling.
- Joint Loss Function: Combines binary classification loss with a custom margin-based loss that pushes synonyms closer and antonyms apart in their respective spaces.

Best Architecture



Best Architecture Results

Dual Encoder

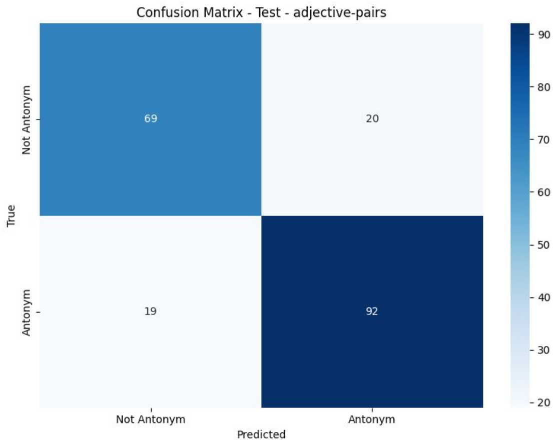
	Precision	Recall	F1-score
Synonyms	**0.92**	**0.88**	**0.90**
Antonyms	**0.90**	**0.93**	**0.91**
Overall Accuracy	**0.91**		
Baseline 1: ICENET	**0.91**		
Baseline 2: Distiller	**0.91**		

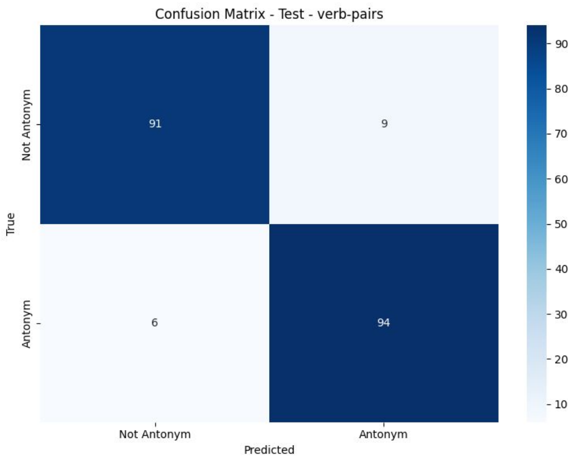
Best Architecture Results

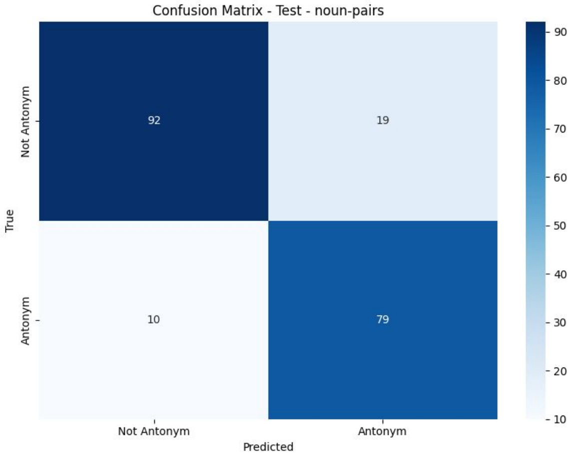
Dual Encoder

	Adjectives	Verbs	Nouns
Dual Encoder	**0.90**	**0.93**	**0.90**
Baseline 1: ICENET	**0.92**	**0.93**	**0.87**
Baseline 2: Distiller	**0.90**	**0.92**	**0.90**

Confusion matrices







Why did dual encoder GCN work?

- Local Interaction Modeling: GCNs can naturally represent each word pair as a small graph and learn how the two words interact — ideal for capturing nuanced relational meaning.
- Context-Aware Learning: By treating each word as a node, GCNs allow the model to consider both individual word features and their relational structure.
- The model has two paths: one learns what makes words similar, the other what makes them opposites.
- It combines both views before passing them through a graph transformer that sees how word pairs relate.
- Synonym pairs should look similar in the "synonym space" — we nudge the model to learn that.
- Antonym pairs should look different in the "antonym space" — we teach the model to recognize contrast.

Limitations

- Fine-tuning BERT could have helped us achieve better accuracy.
- The dataset used has their own biases or limitations in terms of coverage and the nuances of word relationships.
- Did not focus on the different types of antonyms (like gradable or complementary).
- The margin-based loss assumes a clear separation between antonym/synonym spaces, which may not hold for gradable adjectives (e.g., "warm" vs. "hot").

Future Work

- Evaluate the model on larger and more diverse datasets.
- Use attention visualization to trace how graph layers capture relational properties (e.g., symmetry, transitivity) and identify failure modes.
- Explore alternative or enhanced word representations beyond pre-trained embeddings.

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Thank You!

github.com/SamyakSS83/Antonym-detection

Appendix

- Pattern-based neural models leverage lexico-syntactic structures and morphological cues.
- AntSynNET hypothesizes that antonym pairs co-occur in specific syntactic structures (e.g., "either X or Y") more frequently than synonyms.
- It extracts syntactic paths between word pairs from parse trees in sentences and encodes these paths using an LSTM.
- The LSTM outputs a vector representation of the path, fed into a logistic regression classifier to predict antonymy/synonymy.

Appendix

- AntNet extracts dependency paths between word pairs, focusing on both the structure and the morphological features of the words.
- Replaces negated forms with their base forms and adds a feature to indicate negation, improving the model's ability to recognize antonyms formed by morphological changes.
- Encodes the paths using an LSTM and combines this with distributional word vectors.
- Challenges include the requirement of extensive labeled corpora, noise sensitivity, and domain specificity.

Appendix

- Subspace distillation enhances static embeddings by isolating synonymy and antonymy into distinct subspaces to model their unique algebraic properties.
- Synonyms are projected closer in SYN, while antonyms are separated in ANT but linked via trans-transitivity to synonyms.
- Thesauri-Augmented Embeddings combines distributional signals (from corpora) with explicit relational knowledge.
- It injects WordNet antonym pairs into the training data and fine-tunes embeddings using contrastive loss to push antonyms apart while pulling synonyms closer.
- They struggle with words that have multiple senses and do not work for rare or unseen words.